Train Your Own Tiny GPT

Module 7 · Zero to LLM

Shift by one

input `hello worl` → target `ello world` — every position is a mini-exam

Loss & perplexity

- cross-entropy = surprise at the right answer
- random start ≈ **ln(V)** (88 chars → 4.48) — sanity check!
- perplexity = e^loss ≈ "choices it's torn between"

Hiking downhill in fog

part	why
AdamW	momentum + per-parameter step + weight decay
warmup → cosine	small steps, stride, slow down
clip grad 1.0	no cliff jumps
validation	not lost / not memorising
checkpoints	survive crashes

Reading curves

- both ↓ → keep going
- val ↑ while train ↓ → overfitting
- spikes / NaN → LR too high

Next: M08 — fine-tuning